A task-aware framework to quantify MSA-driven effects on phylogenetic inference: a chordate case study of SYK-like proteins

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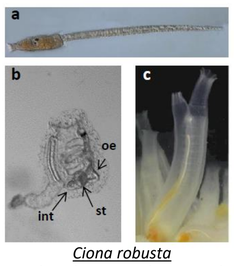
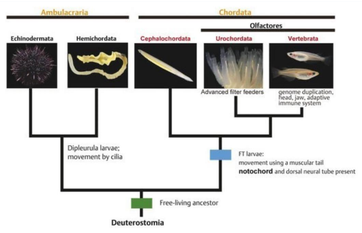
Background and Aim

The tunicate *Ciona robusta*, a chordate with an exclusively innate immune system, offers a pragmatic path to study conservation of key immune molecules in under-represented lineages and to elevate non-canonical organisms to preclinical candidates: high fitness, easy manipulation, and fast life cycle make *C. robusta* attractive for human-oriented translational pipelines. Yet routine studies seldom quantify how multiple sequence alignment (MSA) choices affect downstream phylogenetic inference, especially when annotations are sparse. We address this gap with a minimal protocol to make alignment-to-tree uncertainty measurable and reportable. Our study focuses on tyrosine kinase SYK-like proteins across chordates, starting with the evaluation of *Ciona*'s SYK-like protein sequence using EMBL-EBI PSI-BLAST.

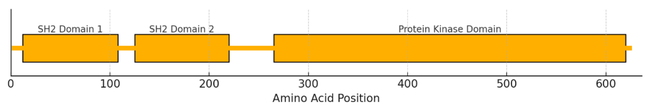
Keywords
Computational phylogenetics, *Ciona robusta*, chordates, MSA, SYK-like



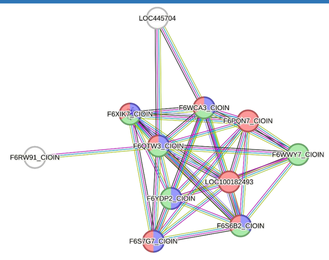
Ciona robusta
a: larval stage; b: juvenile stage; c: adult organism (st, stomach- oe, oesophagus- int, intestine)



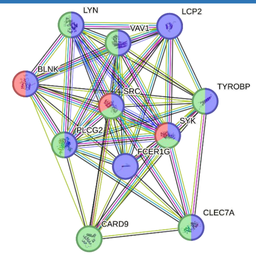
The Evolution of Chordates



Domain Architecture of *Ciona robusta* SYK-like protein



F6QTW3 (SYK-like) *Ciona robusta* PPI
- Immune system process
- Positive regulation of response to stimulus
- SH2 domain



SYK *Homo sapiens* PPI
- Immune response-activating cell surface receptor signaling pathway
- Positive regulation of innate immune response
- SH2 domain binding

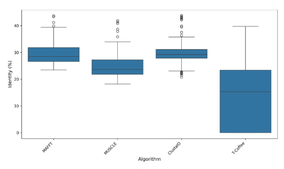
Methods

In a benchmarking across MAFFT, MUSCLE, Clustal Omega, T-Coffee, and ProbCons, algorithm choice measurably shifted the conservation signal: MAFFT and Clustal Omega reached higher pairwise identity (~29–30%) and faster runtimes, but Clustal Omega showed lower sum-of-pairs (SP) score consistency and reduced accuracy in highly divergent regions; MUSCLE averaged ~25% identity with greater gap variability; overall SP-score consistency ranged ~6–9%. After identifying best-performing alignment settings, we adopt a minimal and reproducible design: for each dataset (501-sequence full dataset, 30-sequence subset, and SH2- and kinase-region datasets) we generate two alignments, Untrimmed and Trimmed, and infer one tree per condition using a single robust maximum-likelihood engine, IQ-TREE, with automatic model selection and dual support measures. To ensure interpretability and computational tractability, trees are built on a phylogenetically balanced panel of selected representative taxa (including tunicates, cephalochordates, cyclostomes, ray-finned fish, sauropsids, mammals) from our broader dataset.

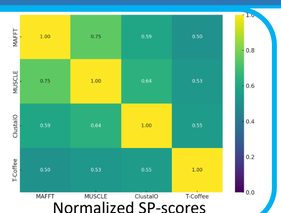
Full_dataset

Comparison	Δ Identity (%)	CI 95%	p-value
MAFFT vs MUSCLE	+4.57	4.13 – 5.01	< 0.0001
MAFFT vs T-Coffee	+15.52	14.55 – 16.45	< 0.0001
MAFFT vs ClustalO	-0.34	-0.77 – 0.09	0.0226
MUSCLE vs ClustalO	-10.95	-5.35 – -4.45	< 0.0001
ClustalO vs T-Coffee	+15.86	14.90 – 16.81	< 0.0001
MUSCLE vs T-Coffee	+10.95	9.99 – 11.91	< 0.0001

Comparison of Identity percentage



Identity % Distribution

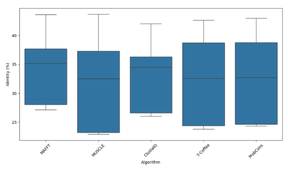


Normalized SP-scores

30-sequence_subset

Comparison	Δ Identity (%)	CI 95%	p-value
MAFFT vs MUSCLE	+2.01	(-1.59 – 5.28)	0.2689
MAFFT vs ClustalO	+0.93	(-2.24 – 4.11)	0.2426
MAFFT vs T-Coffee	+1.23	(-1.84 – 4.28)	0.5965
MAFFT vs ProbCons	+1.06	(-2.16 – 4.35)	0.6074
MUSCLE vs ClustalO	+1.02	(-2.40 – 4.27)	0.6074
MUSCLE vs T-Coffee	+0.77	(-2.44 – 3.84)	0.2496
MUSCLE vs ProbCons	+1.29	(-2.20 – 4.25)	0.2464
ClustalO vs T-Coffee	+0.29	(-2.76 – 3.40)	0.8153
ClustalO vs ProbCons	+0.12	(-2.89 – 3.38)	0.8518
T-Coffee vs ProbCons	+0.47	(-3.20 – 3.65)	0.3586

Comparison of Identity percentage



Identity % Distribution

Normalized SP-scores

SP- and TC- scores for SYK-like domains_dataset

Domain	Method	TC raw	TC %	SP raw	SP %
SH2-1	MUSCLE	0.000	0.00%	0.346	34.63%
SH2-1	ClustalO	0.000	0.00%	0.195	19.50%
SH2-1	T-Coffee	0.051	5.11%	0.188	18.79%
SH2-2	MUSCLE	0.013	1.34%	0.114	11.44%
SH2-2	ClustalO	0.000	0.00%	0.107	10.68%
SH2-2	T-Coffee	0.073	7.28%	0.087	8.66%
Kinase	MUSCLE	0.016	1.63%	0.344	34.44%
Kinase	ClustalO	0.030	3.04%	0.295	29.47%
Kinase	T-Coffee	0.017	1.66%	0.257	25.73%

Note: MAFFT alignments were used as the reference standard for all comparisons.

MSA benchmark

Phylogenetic Inference (MAFFT as reference MSA)

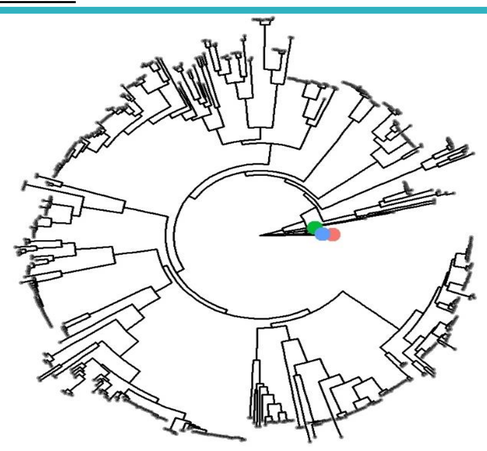
Full_dataset

Condition	Seqs	Sites	Mean gaps / ambiguities (%)	χ^2 composition failures	Best-fit model (BIC)	Best ML logL
UNTRIM	501	6552	86.8	378	JTT+I+R8	−261,981.505
TRIM	501	5608	84.61	309	JTT+R9	−259,229.173

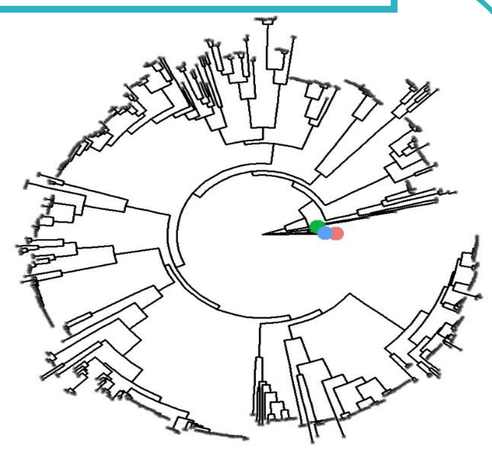
Alignment and model-fit statistics for the full dataset (501 seqs). Trimmed ClipKIT alignment retains 5,608/6,552 sites, slightly reduces mean gaps/ambiguities and χ^2 composition failures, and is fitted by JTT+R9, while the untrimmed matrix is best described by JTT+I+R8.

Metric	Value	Notes
nRF (ML TRIM vs UNTRIM)	0.054	ML trees, 501 taxa
nRF (50% consensus TRIM vs UNTRIM)	0.044	50% consensus trees
Normalized quartet distance (ML)	4.4×10^{-5}	Discordant quartets (normalized)
AU (TRIM, tree 1 / tree 2)	0.571 / 0.429	JTT+R9, zb = 10k
AU (UNTRIM, tree 1 / tree 2)	0.586 / 0.414	JTT+I+R8, zb = 10k

Key tree-comparison metrics for the full dataset (501 seqs)



TRIM-consensus



UNTRIM-consensus

group Ciona SYK-like Human SYK Human ZAP70

Full_dataset tree comparisons confirm that trimmed and untrimmed analyses recover the same deep backbone (nRF ≈ 0.04–0.05; QD_norm ≈ 0), with only minor rearrangements in shallow branches

(preliminary metrics for the other datasets)

30-sequence_subset

Comparison	Seqs	nRF	ΔlnL (UNTRIM - TRIM)	ΔlnL per site
TRIM vs UNTRIM ML trees	30	0.037	+5.9	+0.004

ΔlnL computed by evaluating both ML trees on the UNTRIM alignment (1461 sites) under LG+I+G4.

Condition	Sites	Mean gaps / ambiguities (%)	Seqs > 50% gaps / ambiguities	χ^2 composition failures	Best-fit model (BIC)	Best ML logL
UNTRIM	1461	51.96	18	1	LG+I+G4	-16,515.67
TRIM	641	28.13	7	0	Q.INSECT+G4	-12,212.67

Alignment and model-fit statistics for the 30-sequence subset. ClipKIT trimming reduces alignment length and mean gaps/ambiguities, slightly decreases χ^2 composition failures, and shifts the best-fit model from LG+I+G4 to Q.INSECT+G4.

SP- and TC- scores for SYK-like domains_dataset

Condition	Seqs	Sites	Gaps (%)	Best-fit model (BIC)	Best ML logL
UNTRIM SH2-1	148	156	65.84	LG+G4	-4739.115
TRIM SH2-1	148	97	46.70	LG+G4	-4432.759
UNTRIM SH2-2	189	96	22.73	Q.INSECT+G4	-7234.104
TRIM SH2-2	189	83	12.06	Q.INSECT+G4	-7145.467

Alignment characteristics and best-fit substitution models for UNTRIM and TRIM SH2-1 and SH2-2 domain datasets (number of sequences, sites, gap content, and ML log-likelihood).

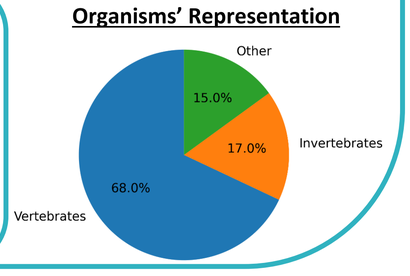
Condition	Sites	Gaps (%)	Best-fit model (BIC)	Best ML logL
UNTRIM Kinase	362	36.7	Q.INSECT+I+R7	-39,520.7
TRIM Kinase	312	29.7	Q.INSECT+R7	-39,202.2

Kinase-domain analyses: trimming slightly reduces alignment length and gaps, with nRF = 0.169 quantifying its topological impact across 501 seqs

Domain	Seqs (with domain)	nRF (TRIM vs UNTRIM)
Kinase	501	0.169
SH2-1	148	0.297
SH2-2	189	0.199

Normalized RF distances between TRIM and UNTRIM ML trees for the kinase and SH2 domains, indicating a stronger topological impact of trimming on SH2-1 and SH2-2 (nRF ≈ 0.30 and 0.20) than on full-length kinase (nRF ≈ 0.17).

Organisms' Representation



Results and Conclusions

- On the full dataset, ClipKIT kept 5,608/6,552 sites, ModelFinder selected JTT+I+R8 and JTT+R9 as best-fit models for UNTRIM and TRIM, respectively, supported by ultrafast bootstrap (UFBoot) and Shimodaira-Hasegawa-like approximate likelihood-ratio test (SH-aLRT). Trimming effects were quantified via Robinson–Foulds (RF) and quartet distances. Preliminary results suggest that Trimmed vs Untrimmed consensus trees are highly congruent (preliminary normalized RF ≈ 0.080 under a fixed LG+R9 model, decreasing to ≈ 0.04–0.05 after re-optimizing models with ModelFinder): trimming mainly prunes low-information sites while preserving the deep backbone. Support patterns are concordant on principal clades and model choice chiefly affects branch lengths.

- Applied to SYK-like families, this design reduces topology dispersion relative to unfiltered alignments and clarifies focal relationships (tunicates vs vertebrates) without inflating spurious high-support clades. It delivers a practical recipe that helps in converging on alignment strategies fit for purpose, while communicating residual uncertainty in the comparative immunology of chordates.

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